GALAMAD AEROSPACE

GFSK Transceiver

Doppler Robustness — Lab Test Guide

Testing TX/RX resilience to orbital Doppler from a stationary bench

Field	Value
Document	Doppler Robustness Lab Test Guide
Applies to	GALAMAD GFSK Transceiver web app (gfsk_ui.py) + doppler.py
Link under test	437.000 MHz, 200 kbps, ±50 kHz dev, GFSK BT=0.5, sync 0x2DD4
Hardware	HackRF (TX), RTL-SDR / HackRF (RX), Nucleo G474RE + Si443x (flight radio)
Version	1.0
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Why this guide exists

Orbital velocities (up to ~7.5 km/s) cannot be reproduced by moving the radios. Instead we shift and sweep the **carrier** electronically — the only thing Doppler does to a non-spread GFSK link — and measure whether the receiver still decodes. This guide covers the four tests, the exact app controls, and pass criteria.

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1. Purpose & scope

This guide describes how to characterise and demonstrate the robustness of the GALAMAD GFSK uplink/downlink in the presence of the Doppler shift and Doppler rate a low-Earth-orbit satellite imposes. It applies to:

- the HackRF → SDR ground link (software modem), and
- the Nucleo + Si443x flight radio's receiver (hardware AFC/clock recovery).

The objective is to answer two questions with numbers: **(1)** how much static frequency error can the receiver tolerate before decoding fails, and **(2)** how fast a carrier slew (Hz/s) can it track before failing. Both are compared against the peak values a real pass produces at 437 MHz.

2. Background: what Doppler does to the link

For a narrowband, non-spread signal, satellite motion appears entirely as a time-varying carrier frequency offset:

```
f_observed(t) = f_carrier &times; (1 &minus; v_r(t)/c)      [v_r = range-rate, +ve receding]
Doppler(t)      = &minus; f_carrier &times; v_r(t)/c
Doppler rate(t) = &minus; f_carrier &times; a_r(t)/c        [a_r = range acceleration]
```

Doppler is therefore separable into two effects that stress the receiver differently:

- **Offset** — a static shift (worst near AOS/LOS). At 437 MHz this reaches roughly **±8–10 kHz** for LEO.
- **Rate** — the carrier *slewing*. This is what stresses AFC/tracking loops: peak values are on the order of **tens to ~100 Hz/s** at 437 MHz.

> **Key insight**
>
> A static offset tests the receiver's **acquisition window**; only a **moving** carrier tests its **tracking**. The app reproduces both by injecting the shift/slew into the transmitted IQ.

3. Prerequisites & bench setup

- Web app running (from the canonical symlink path):

```
cd "/home/luwangac/Downloads/PROSat Designs/SDR Apps/GALAMAD-GFSK-Transceiver"
./start.sh            # then open http://localhost:5000
```

- HackRF connected for TX; RTL-SDR or a second HackRF for RX. Select both in the app.
- RX gain: LNA 24 / VGA 48 for a HackRF–HackRF bench; LNA 0 / VGA 8 for the very strong Si443x beacon.
- TX power: tx_gain 40 with amp ON (use the *Bench* preset); the link is freq-independent, so both radios follow the frequency boxes.
- For the flight-radio test: Nucleo flashed with the alternating firmware and the Si443x card showing **AUTO RX/TX**.
- Set the app **freq** and **rx freq** to **437.000 MHz** (the Si443x itself is programmed near 433.997 MHz; use 434.000 when listening to the beacon).

4. The Satellite Doppler card

The full-width **SATELLITE DOPPLER** card is the control surface and the readout. It has three parts: *Satellite & Observer*, *Pass Prediction & RX Tracking*, and *Lab Doppler stress*.

4.1 Load a satellite

- Choose a Celestrak **group** (stations, amateur, weather, cubesat, …) or type a **NORAD ID** and press **Fetch TLE**. Elements are requested as OMM/JSON, so new 6-digit IDs (e.g. 100796) work; if Celestrak has none, the app falls back to the GALAMAD tracker (launchedsats.galamad.space).
- You can also paste a TLE or an OMM JSON block and press **Use pasted TLE**.

4.2 Set the observer

- **Use my location** fills lat/lon/alt from the browser (needs localhost or https).
- **GALAMAD Studio** snaps to 1.34413 °N, 103.88648 °E (the default).
- Or type the coordinates and press **Apply observer**.

4.3 Read the numbers you must reproduce

- Set the **Downlink** frequency (MHz) and **Min elevation**, then **Predict next pass**. The plot is Doppler (Hz) vs time across the pass.
- The live line shows the current **elevation, azimuth, range, range-rate, Doppler (Hz / ppm) and Doppler rate (Hz/s)** for the tracked satellite. **Note the peak |Doppler| and peak |rate|** — these are your Test A and Test B targets.
- **Track RX** retunes the receiver to follow the predicted downlink (mitigation under test).

5. Test A — Static offset tolerance

Measures: the receiver's frequency-acquisition window / AFC pull-in. Emulates a frozen Doppler (a single moment of a pass).

Procedure

- RX at 437.000 MHz; leave **Track RX OFF**.
- In *Lab Doppler stress*, set **Rate Hz/s = 0** and **Offset Hz = 0**; press **Apply to TX**, then **Send**. Capture & Decode → record the decoded payload / CRC-valid count (baseline).
- Repeat for offset = ±500, ±1000, ±2000, ±5000, ±8000, ±10000 Hz. Press **Send** after each change.
- Tabulate CRC-valid decode vs injected offset.

Equivalent alternative

Instead of injecting on TX you may offset the **RX** frequency box by the same amount; this additionally exercises the receiver's tuning accuracy. Pick one method and be consistent.

6. Test B — Doppler rate robustness

Measures: how fast a carrier slew the receiver can acquire and track. This is the decisive test — a pass fails only if the rate exceeds tolerance.

Procedure

- Leave **Track RX OFF**. Set **Offset Hz = 0**.
- Set **Rate Hz/s** to 10, 25, 50, 100, 200 Hz/s in turn. Press **Apply to TX** then **Send**; Capture & Decode after each.
- The carrier slews continuously; each loop sweeps (rate × loop duration) Hz. Use **Use predicted** to load the exact rate from the live pass.
- Record decode success vs rate; the highest rate that still yields CRC-valid packets is the tracking limit.

Interpreting the sweep

The generated file loops, so the sweep restarts each pass. The phase discontinuity falls inside the silent inter-packet gap, so it does not disturb the receiver.

7. Test C — Verify the mitigation (RX tracking)

Measures: that the software Doppler compensation actually cancels the shift.

- Load a satellite and observer; confirm the live Doppler in the card is non-zero.
- Enable **Track RX**. The status shows *RX tuned <f> MHz*; verify it moves as the prediction changes.
- For a closed-loop demonstration, inject a **static** offset on TX while Track RX is on and configured for the same pass; the receiver re-centres on the slewed carrier and decodes.
- Compare the decode-rate limit from Test B with tracking on vs off — the improvement quantifies the benefit of compensation.

8. Test D — SST01 / Si443x flight-radio AFC

Measures: the flight radio's own AFC/clock-recovery margin, with no software help.

- Flash the alternating firmware; confirm the Si443x card reads **AUTO RX/TX** and the log shows *Si443x TX* / *Si443x RX* lines.
- Drive the HackRF (UI **Send**) at 434.000 MHz so the Si443x RX window receives it.
- Inject **Offset Hz** = the model's peak |Doppler| (e.g. ±9 kHz); watch the **Si443x Transceiver Log** for *PAYLOAD:* lines. Sweep the offset to find where the Si443x stops decoding — its AFC pull-in.
- Repeat with a **Rate Hz/s** sweep to test whether its AFC tracks a moving carrier.

9. Recording results & pass criteria

Test	Sweep	Metric	Pass criterion
A	Static offset ±0–10 kHz	Max offset with CRC-valid decode	≥ peak |Doppler| of target pass
B	Rate 0–200 Hz/s	Max rate with CRC-valid decode	≥ peak |Doppler rate| of target pass
C	Tracking on/off	Rate limit improvement	Tracking restores margin at peak rate
D	Offset / rate (Si443x)	Max offset/rate with PAYLOAD	≥ peak values for the mission orbit

Green light

The link is considered Doppler-robust when, for the worst-case pass of the mission orbit, both the peak offset and the peak rate are decoded with CRC valid — either open-loop (radio tolerance) or with RX tracking enabled.

10. Limitations

- **Ideal channel.** Doppler is injected at baseband; there is no range-delay, multipath or path-loss model. This is correct for isolating Doppler (a non-spread GFSK PHY sees it purely as carrier offset + rate).
- **Loop length.** A constant rate reproduces the steep segment of a pass, not the whole S-curve. For a full pass, extend the file length or request the predicted-curve IQ mode.
- **Altitude.** Browser geolocation altitude is coarse; it has negligible effect on topocentric range-rate at these ranges and may be left at the default.
- **Element age.** SGP4 accuracy degrades with TLE age; for precise passes use elements less than ~1 day old.

Appendix A — Controls & endpoints

Control / endpoint	Purpose
Fetch TLE / group / NORAD ID	Load elements (Celestrak OMM, GALAMAD fallback)
Use my location / GALAMAD Studio	Set the observer for topocentric geometry
Predict next pass	AOS/TCA/LOS, max elevation, max Doppler, curve
Track RX	Retune the receiver to follow the predicted downlink
Lab Doppler stress: Offset Hz	Constant carrier offset injected into TX IQ
Lab Doppler stress: Rate Hz/s	Linear carrier slew injected into TX IQ
Apply to TX / Use predicted / Clear	Set, auto-fill from live pass, or zero the injection
POST /doppler/tle	Load elements (group | catnr | text)
POST /doppler/observer	Set lat/lon/alt
POST /doppler/config	Downlink frequency, elevation mask
POST /doppler/inject	Set TX Doppler offset & rate (off_hz, rate_hz_s)
GET /doppler/status	Live look, Doppler, Doppler rate, inject state
GET /doppler/predict	Next-pass prediction samples

Appendix B — Formulas

```
Doppler shift      f_d = &minus; f_c &middot; v_r / c           (v_r = line-of-sight range-rate)
Doppler rate        f_d' = &minus; f_c &middot; a_r / c        (a_r = range acceleration)
Injected phase     &phi;(t) = 2&pi;( f_0&middot;t + &frac12;&middot;k&middot;t&sup2; )   (f_0 = offset,
Per-loop sweep     &Delta;f = k &middot; (loop duration)
```

Constants: c = 299 792 458 m/s. WGS-84 ellipsoid for observer/topocentric geometry. SGP4 propagation of the loaded element set.

— end —